

X-ray), geo-scientific collaboration, collaborative design in CAD modelling, collaborative archaeology and remote teaching of physical activities/skills (such as in rehabilitation and dance performance).

**In collaborative work.** Tele-immersion will enable groups of engineers to work closely together across great distances on computerised designs, for say, developing innovative equipment (Fig. 5).

While different teams of engineers might be thousands of kilometres away, they would be able to work together on the same project as if they were sitting together on the same workbench.

Archaeologists from around the globe will have the same sort of experience during a critical excavation. Rarefied experts in building inspection or engine repair will be able to visit various sites without incurring time and money expenditure on air travel. Similarly, collaboration will be



Fig. 5: A collaborative design tele-immersion session in progress (Source: [www.dvetelepresence.com](http://www.dvetelepresence.com))

possible while working on mechanical CAD. A team of designers will be able to work in partnership from remote sites in an interactive design process. They will be able to influence a virtual model, right from the conceptual to the production stage.

### What the future holds

One of the current research areas is how to make tele-immersion more natural, perhaps by doing away with the headgear and glasses. Efforts are on for making the tele-immersion experience possible by simply entering a tele-cubicle. One solution is to use a screen that transmits diverse information to each eye, using swivelling

pixels that track either left or right eye.

Another probability is to turn the entire tele-immersion room onto a screen. Walls, chairs, roof, etc could be coated with special light-sensitive material.

A camera would photograph surfaces, computers would calculate their shapes in 3D and projectors would screen pre-warped images, making it appear as if these filled the room.

In years to come, perhaps one of your office walls will sparkle and your teammate will be sitting right in front of you, as if in person. On the other side, your teammate will enjoy the same experience. This possibility will make air travel quite rare. This is definitely going to have a favourable impact on our environment too with less greenhouse gas emission, which will make this planet a better place to live on. ●

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